Compare physical and chemical changes and identify indicators of energy change in chemical reactions

Learning intention

- compare physical and chemical changes and identify indicators of energy change in chemical reactions.

Teach and model

- examining how the physical and chemical properties of a substance will affect its production or use.
- investigating and identifying energy changes in different chemical reactions such as differences in temperature.
- discussing where indicators of chemical change are used for identifying the presence of particular substances, such as in soil, water and medical testing kits.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.